



# Linear offset-free Model Predictive Control<sup>☆</sup>

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## ABSTRACT

This work addresses the problem of offset-free Model Predictive Control (MPC) when tracking an asymptotically constant reference. In the first part, compact and intuitive conditions for offset-free MPC control are introduced by using the arguments of the internal model principle. In the second part, we study the case where the number of measured variables is larger than the number of tracked variables. The plant model is augmented only by as many states as there are tracked variables, and an algorithm which guarantees offset-free tracking is presented. In the last part, offset-free tracking properties for special implementations of MPC schemes are briefly discussed.

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## 1. Introduction

The main concept of Model Predictive Control (MPC) is to use a *model* of the plant to *predict* the future evolution of the system (Borrelli, Bemporad, Fodor, & Hrovat, 2006; Garcia, Prett, & Morari, 1989; Hrovat, 1996; Mayne, Rawlings, Rao, & Scokaert, 2000; Morari & Lee, 1999). At each time step  $t$  a certain performance index is optimized over a sequence of future input moves subject to operating constraints. The first of such optimal moves is the *control* action applied to the plant at time  $t$ . At time  $t+1$ , a new optimization is solved over a shifted prediction horizon.

In order to obtain offset-free control with MPC, the system model is augmented with a disturbance model which is used to estimate and predict the mismatch between measured and predicted outputs. The state and disturbance estimates are used to initialize the MPC problem. The MPC algorithms presented in Badgwell and Muske (2002), Liu and Brosilow (1987), Maciejowski (1998), Magni, De Nicolao, and Scattolin (2001), Meadowcroft, Stephanopoulos, and Brosilow (1992), Pannocchia (2003), Pannocchia and Bemporad (2007), Pannocchia and Rawlings (2003) and Zafiriou and Morari (1988) guarantee offset-free control when no constraints are active. They differ in the MPC problem setup, the

type of disturbance model used and the assumptions which guarantee offset-free control.

This work addresses the problem of offset-free MPC when tracking a constant reference and using linear system models. It represents a thorough study of the main conditions and design algorithms which guarantee offset-free MPC for a wide range of practically relevant cases.

We distinguish between the number  $p$  of measured outputs, the number  $r$  of outputs which one desires to track (called “tracked outputs”), and the number  $n_d$  of disturbances. We divide the paper into three parts. The first part builds on the work in Pannocchia (2003); Pannocchia and Rawlings (2003) and summarizes, in a compact and intuitive manner, the conditions that need to be satisfied to obtain offset-free MPC by using the arguments of the internal model principle. A simple proof of zero steady-state offset is provided when  $n_d = r = p$ .

The second part considers the case where the number of measured variables  $p$  is greater than the number of tracked variables  $r$ . In this case, the approaches in Pannocchia and Rawlings (2003) and Muske and Badgwell (2002) allow one to freely choose the disturbance models, but they require the number of added disturbance states  $n_d$  to be at least equal to the number of *measured variables*  $p$ , rather than *tracked variables*  $r$ . Our contribution is twofold. First, a simple algorithm for computing the space spanned by the offset is provided when  $n_d < p$ . Second, we show how to construct a controller/observer combination such that zero offset is achieved when the plant model is augmented only by as many additional states as there are tracked variables ( $n_d = r < p$ ), yielding an MPC with minimal complexity. This effect is particularly useful in the area of explicit Model Predictive Control (Bemporad, Morari, Dua,

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& Pistikopoulos, 2000), where the number of parameters affects the complexity of the problem.

In the last part, we provide insights on zero steady-state offset when one/infinity norm objective functions,  $\delta u$  formulation and explicit MPC are implemented. We conclude with two illustrative examples.

## 2. Preliminaries

In this section, we present the standard MPC design flow: the model choice, the observer design and the controller design. Consider the discrete-time time-invariant system

$$\begin{cases} x_m(t+1) = f(x_m(t), u(t)) \\ y_m(t) = g(x_m(t)) \\ z(t) = Hy_m(t) \end{cases} \quad (1)$$

with the constraints

$$Ex_m(t) + Lu(t) \leq M. \quad (2)$$

In (1)–(2),  $x_m(t) \in \mathbb{R}^n$ ,  $u(t) \in \mathbb{R}^m$  and  $y_m(t) \in \mathbb{R}^p$  are the state, input, measured output vector, respectively. The controlled variables  $z(t) \in \mathbb{R}^r$  are a linear combination of the measured variables. Without any loss of generality we assume  $H$  to have full row rank. The matrices  $E$ ,  $L$  and  $M$  define state and input constraints.

The objective is to design an MPC (Garcia et al., 1989; Mayne et al., 2000) based on a linear system model of (1) in order to have  $z(t)$  track  $r(t)$ , where  $r(t) \in \mathbb{R}^p$  is the reference signal, which we assume to converge to a constant, i.e.  $r(t) \rightarrow r_\infty$  as  $t \rightarrow \infty$ . Moreover, we require zero steady-state tracking error, i.e.,  $(z(t) - r(t)) \rightarrow 0$  for  $t \rightarrow \infty$ .

### 2.1. The plant model

The MPC scheme will make use of the following linear time-invariant system model of (1):

$$\begin{cases} x(t+1) = Ax(t) + Bu(t) \\ y(t) = Cx(t), \end{cases} \quad (3)$$

where  $x(t) \in \mathbb{R}^n$ ,  $u(t) \in \mathbb{R}^m$  and  $y(t) \in \mathbb{R}^p$  are the state, input and output vector, respectively. We assume that the pair  $(A, B)$  is controllable, and the pair  $(C, A)$  is observable. Furthermore,  $C$  is assumed to have full row rank.

### 2.2. The observer design

The plant model (3) is augmented with a disturbance model in order to capture the mismatch between (1) and (3) in steady state. Several disturbance models have been presented in the literature (Badgwell & Muske, 2002; Liu & Brosilow, 1987; Meadowcroft et al., 1992; Pannocchia, 2003; Pannocchia & Rawlings, 2003; Zafiriou & Morari, 1988). In this note we follow (Pannocchia & Rawlings, 2003) and use the form:

$$\begin{cases} x(t+1) = Ax(t) + Bu(t) + B_d d(t) \\ d(t+1) = d(t) \\ y(t) = Cx(t) + C_d d(t) \end{cases} \quad (4)$$

with  $d(t) \in \mathbb{R}^{n_d}$ . With abuse of notation we have used the same symbols for state and outputs of system (3) and system (4). Later we will focus on specific versions of the model (4).

The observer estimates both states and disturbances based on this augmented model. Conditions for the observability of (4) are given in the following proposition.

**Proposition 1** (Badgwell & Muske, 2002; Morari & Stephanopoulos, 1980a; Pannocchia & Rawlings, 2003). The augmented system (4) is observable if and only if  $(C, A)$  is observable and

$$\begin{bmatrix} A - I & B_d \\ C & C_d \end{bmatrix} \text{ has full column rank.} \quad (5)$$

**Proof.** From the Hautus observability condition system (4) is observable iff

$$\begin{bmatrix} A^T - \lambda I & 0 & C^T \\ B_d^T & I - \lambda I & C_d^T \end{bmatrix} \text{ has full row rank } \forall \lambda. \quad (6)$$

Again from the Hautus condition, the first set of rows is linearly independent iff  $(C, A)$  is observable. The second set of rows is linearly independent from the first  $n$  rows except possibly for  $\lambda = 1$ . Thus, for the augmented system the Hautus condition needs to be checked for  $\lambda = 1$  only, where it becomes (5).  $\square$

**Remark 1.** Note that for condition (5) to be satisfied the number of disturbances in  $d$  needs to be smaller or equal to the number of available measurements in  $y$ ,  $n_d \leq p$ . Condition (5) can be nicely interpreted. It requires that the model of the disturbance effect on the output  $d \rightarrow y$  must not have a zero at  $(1, 0)$ . Alternatively we can look at the steady state of system (4)

$$\begin{bmatrix} A - I & B_d \\ C & C_d \end{bmatrix} \begin{bmatrix} x_\infty \\ d_\infty \end{bmatrix} = \begin{bmatrix} 0 \\ y_\infty \end{bmatrix} \quad (7)$$

where we have denoted the steady state values with a subscript  $\infty$  and have omitted the forcing term  $u$  for simplicity. We note that from the observability condition (5) for system (4) Eq. (7) is required to have a unique solution, which means, that we must be able to deduce a unique value for the disturbance  $d_\infty$  from a measurement of  $y_\infty$  in steady state.

The state and disturbance estimator is designed based on the augmented model as follows:

$$\begin{aligned} \begin{bmatrix} \hat{x}(t+1) \\ \hat{d}(t+1) \end{bmatrix} &= \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}(t) \\ \hat{d}(t) \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u(t) \\ &+ \begin{bmatrix} L_x \\ L_d \end{bmatrix} (-y_m(t) + C\hat{x}(t) + C_d\hat{d}(t)) \end{aligned} \quad (8)$$

where  $L_x$  and  $L_d$  are chosen so that the estimator is stable. We remark that the results of this paper are independent on the choice of the method for computing  $L_x$  and  $L_d$ . We then have the following property.

**Proposition 2.** Suppose the observer (8) is stable. Then,  $\text{rank}(L_d) = n_d$ .

**Proof.** From (8) it follows

$$\begin{aligned} \begin{bmatrix} \hat{x}(t+1) \\ \hat{d}(t+1) \end{bmatrix} &= \begin{bmatrix} A + L_x C & B_d + L_x C_d \\ L_d C & I + L_d C_d \end{bmatrix} \begin{bmatrix} \hat{x}(t) \\ \hat{d}(t) \end{bmatrix} \\ &+ \begin{bmatrix} B \\ 0 \end{bmatrix} u(t) - \begin{bmatrix} L_x \\ L_d \end{bmatrix} y_m(t). \end{aligned} \quad (9)$$

By stability, the observer has no poles at  $(1, 0)$  and therefore

$$\det \left( \begin{bmatrix} A - I + L_x C & B_d + L_x C_d \\ L_d C & L_d C_d \end{bmatrix} \right) \neq 0. \quad (10)$$

For Eq. (10) to hold, the last  $n_d$  rows of the matrix have to be of full row rank. A necessary condition is that  $L_d$  has full row rank.  $\square$

In the rest of this section, we will focus on the case  $n_d = p$ .

**Proposition 3.** Suppose the observer (8) is stable. Choose  $n_d = p$ . The steady state of the observer (8) satisfies:

$$\begin{bmatrix} A - I & B \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_\infty \\ u_\infty \end{bmatrix} = \begin{bmatrix} -B_d \hat{d}_\infty \\ y_{m,\infty} - C_d \hat{d}_\infty \end{bmatrix} \quad (11)$$

where  $y_{m,\infty}$  and  $u_\infty$  are the steady state measured output and input of the system (1),  $\hat{x}_\infty$  and  $\hat{d}_\infty$  are state and disturbance estimates from the observer (8) at steady state, respectively.

**Proof.** From (8) we note that the disturbance estimate  $\hat{d}$  converges only if  $L_d(-y_{m,\infty} + C\hat{x}_\infty + C_d\hat{d}_\infty) = 0$ . As  $L_d$  is square by assumption and nonsingular by Proposition 2 this implies that at steady state, the observer estimates (8) satisfy

$$-y_{m,\infty} + C\hat{x}_\infty + C_d\hat{d}_\infty = 0. \quad (12)$$

Eq. (11) follows directly from (12) and (8).  $\square$

Next we particularize the conditions in Proposition 1 to special disturbance classes. The results will be useful later in this paper. The following two corollaries follow directly from Proposition 1.

**Corollary 1.** The augmented system (4) with  $n_d = p$  and  $C_d = I$  is observable if and only if  $(C, A)$  is observable and

$$\det \begin{bmatrix} A - I & B_d \\ C & I \end{bmatrix} = \det(A - I - B_d C) \neq 0. \quad (13)$$

**Remark 2.** We note here clearly how the observability requirement restricts the choice of the disturbance model. If the plant has no integrators, then  $\det(A - I) \neq 0$  and we can choose  $B_d = 0$ . If the plant has integrators then  $B_d$  has to be chosen specifically to make  $\det(A - I - B_d C) \neq 0$ .

### 2.3. The MPC design

Denote by  $z_\infty = Hy_{m,\infty}$  and  $r_\infty$  the tracked measured outputs and their references at steady state, respectively. For offset-free tracking at steady state we want  $z_\infty = r_\infty$ . The observer condition (11) suggests that at steady state the MPC should satisfy

$$\begin{bmatrix} A - I & B \\ HC & 0 \end{bmatrix} \begin{bmatrix} x_\infty \\ u_\infty \end{bmatrix} = \begin{bmatrix} -B_d \hat{d}_\infty \\ r_\infty - HC_d \hat{d}_\infty \end{bmatrix} \quad (14)$$

where  $x_\infty$  is the controller state at steady state. For  $x_\infty$  and  $u_\infty$  to exist for any  $\hat{d}_\infty$  and  $r_\infty$  the matrix  $\begin{bmatrix} A - I & B \\ HC & 0 \end{bmatrix}$  must be of full row rank which implies  $m \geq r$ .

The MPC is designed as follows

$$\begin{aligned} \min_{u_0, \dots, u_{N-1}} \quad & \|x_N - \bar{x}_t\|_P^2 + \sum_{k=0}^{N-1} \|x_k - \bar{x}_t\|_Q^2 + \|u_k - \bar{u}_t\|_R^2 \\ \text{subj. to} \quad & Ex_k + Lu_k \leq M, \quad k = 0, \dots, N \\ & x_{k+1} = Ax_k + Bu_k + B_d d_k, \quad k = 0, \dots, N \\ & d_{k+1} = d_k, \quad k = 0, \dots, N \\ & x_0 = \hat{x}(t) \\ & d_0 = \hat{d}(t), \end{aligned} \quad (15)$$

with  $\bar{u}_t$  and  $\bar{x}_t$  given by

$$\begin{bmatrix} A - I & B \\ HC & 0 \end{bmatrix} \begin{bmatrix} \bar{x}_t \\ \bar{u}_t \end{bmatrix} = \begin{bmatrix} -B_d \hat{d}(t) \\ r(t) - HC_d \hat{d}(t) \end{bmatrix} \quad (16)$$

and where  $\|x\|_M^2 \triangleq x^T M x$ ,  $Q \geq 0$ ,  $R > 0$ , and  $P$  satisfies the Riccati equation

$$P = A^T P A - (A^T P B)(B^T P B + R)^{-1} (B^T P A) + Q. \quad (17)$$

Note that we distinguish between the *current* input  $u(t)$  to system (3) at time  $t$ , and the optimization variables  $u_k$  in the optimization problem (15). Analogously,  $x(t)$  denotes the system state at time  $t$ , while the variable  $x_k$  denotes the *predicted* state at time  $t + k$  obtained by starting from the state  $x_0 = x(t)$  and applying to system (3) the input sequence  $u_0, \dots, u_{k-1}$ .

Let  $U^*(t) = \{u_0^*, \dots, u_{N-1}^*\}$  be the optimal solution of (15)–(16) at time  $t$ . Then, the first sample of  $U^*(t)$  is applied to system (1)

$$u(t) = u_0^*. \quad (18)$$

Denote by  $c_0(\hat{x}(t), \hat{d}(t), y_{ref}) = u_0^*(\hat{x}(t), \hat{d}(t), r(t))$  the control law when the estimated state and disturbance are  $\hat{x}(t)$  and  $\hat{d}(t)$ , respectively. Then the closed loop system obtained by controlling (1) with the MPC (15)–(16)–(18) and the observer (8) is:

$$\begin{aligned} x(t+1) &= f(x(t), c_0(\hat{x}(t), \hat{d}(t), r(t))) \\ \hat{x}(t+1) &= (A + L_x C)\hat{x}(t) + (B_d + L_x C_d)\hat{d}(t) \\ &\quad + Bc_0(\hat{x}(t), \hat{d}(t), r(t)) - L_x y_m(t) \\ \hat{d}(t+1) &= L_d C\hat{x}(t) + (I + L_d C_d)\hat{d}(t) - L_d y_m(t). \end{aligned} \quad (19)$$

### 3. Number of disturbance states $n_d$ equal to number of measured outputs $p$

Often in practice, one desires to track all measured outputs with zero offset. Choosing  $n_d = p = r$  is thus a natural choice. Such disturbance models have already been shown to yield offset-free control (Badgwell & Muske, 2002; Muske & Badgwell, 2002; Pannocchia & Rawlings, 2003). Such zero-offset property continues to hold if only a subset of the measured outputs are to be tracked, i.e.,  $n_d = p > r$ .

Next we provide a very simple proof for offset-free control when  $n_d = p$  (Badgwell & Muske, 2002; Muske & Badgwell, 2002; Pannocchia & Rawlings, 2003).

**Theorem 1.** Consider the case  $n_d = p$ . Assume that for  $r(t) \rightarrow r_\infty$  as  $t \rightarrow \infty$ , the MPC problem (15)–(16) is feasible for all  $t \in \mathbb{N}^+$ , unconstrained for  $t \geq j$  with  $j \in \mathbb{N}^+$  and the closed-loop system (19) converges to  $\hat{x}_\infty, \hat{d}_\infty, y_{m,\infty}$ , i.e.,  $\hat{x}(t) \rightarrow \hat{x}_\infty, \hat{d}(t) \rightarrow \hat{d}_\infty, y_m(t) \rightarrow y_{m,\infty}$  as  $t \rightarrow \infty$ . Then  $z(t) = Hy_m(t) \rightarrow r_\infty$  as  $t \rightarrow \infty$ .

**Proof.** Consider the the MPC problem (15)–(16). At steady state  $u(t) \rightarrow u_\infty = c_0(\hat{x}_\infty, \hat{d}_\infty, r_\infty)$ ,  $\bar{x}_t \rightarrow \bar{x}_\infty$  and  $\bar{u}_t \rightarrow \bar{u}_\infty$ . Note that the steady state controller input  $u_\infty$  (computed and implemented) might be different from the steady state target input  $\bar{u}_\infty$ .

The asymptotic values  $\hat{x}_\infty, \bar{x}_\infty, u_\infty$  and  $\bar{u}_\infty$  satisfy the observer conditions (11)

$$\begin{bmatrix} A - I & B \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_\infty \\ u_\infty \end{bmatrix} = \begin{bmatrix} -B_d \hat{d}_\infty \\ y_{m,\infty} - C_d \hat{d}_\infty \end{bmatrix} \quad (20)$$

and the controller requirement (16)

$$\begin{bmatrix} A - I & B \\ HC & 0 \end{bmatrix} \begin{bmatrix} \bar{x}_\infty \\ \bar{u}_\infty \end{bmatrix} = \begin{bmatrix} -B_d \hat{d}_\infty \\ r_\infty - HC_d \hat{d}_\infty \end{bmatrix}. \quad (21)$$

Define  $\delta x = \hat{x}_\infty - \bar{x}_\infty$ ,  $\delta u = u_\infty - \bar{u}_\infty$  and the offset  $\epsilon = z_\infty - r_\infty$ . Notice that the steady state target values  $\bar{x}_\infty$  and  $\bar{u}_\infty$  are both functions of  $r_\infty$  and  $\hat{d}_\infty$  as given by (21). Left multiplying the second row of (20) by  $H$  and subtracting (21) from the result, we obtain

$$\begin{aligned} (A - I)\delta x + B\delta u &= 0 \\ HC\delta x &= \epsilon. \end{aligned} \quad (22)$$

Next we prove that  $\delta x = 0$  and thus  $\epsilon = 0$ .

Consider the MPC problem (15)–(16) and the following change of variables  $\delta x_k = x_k - \bar{x}_t$ ,  $\delta u_k = u_k - \bar{u}_t$ . Notice that  $H y_k - r(t) = H C x_k + H C_d d_k - r(t) = H C \delta x_k + H C \bar{x}_t + H C_d d_k - r(t) = H C \delta x_k$  from condition (16) with  $\hat{d}(t) = d_k$ . Similarly, one can show that  $\delta x_{k+1} = A \delta x_k + B \delta u_k$ . Then, the MPC problem (15) becomes:

$$\begin{aligned} \min_{\delta u_0, \dots, \delta u_{N-1}} \quad & \|\delta x_N\|_P^2 + \sum_{k=0}^{N-1} \|\delta x_k\|_Q^2 + \|\delta u_k\|_R^2 \\ \text{subj. to} \quad & E \delta x_k + L \delta u_k \leq M_2, \quad k = 0, \dots, N \\ & \delta x_{k+1} = A \delta x_k + B \delta u_k, \quad k = 0, \dots, N \\ & \delta x_0 = \delta x(t), \\ & \delta x(t) = \hat{x}(t) - \bar{x}_t. \end{aligned} \quad (23)$$

Denote by  $K_{\text{MPC}}$  the unconstrained MPC controller (23), i.e.,  $\delta u_0^* = K_{\text{MPC}} \delta x(t)$ . At steady state  $\delta u_0^* \rightarrow u_\infty - \bar{u}_\infty = \delta u$  and  $\delta x(t) \rightarrow \hat{x}_\infty - \bar{x}_\infty = \delta x$ . Therefore, at steady state,  $\delta u = K_{\text{MPC}} \delta x$ . From (22)

$$(A - I + B K_{\text{MPC}}) \delta x = 0. \quad (24)$$

Since  $P$  satisfies the Riccati equation (17),  $K_{\text{MPC}}$  is a stabilizing control law, which implies  $(A - I + B K_{\text{MPC}})$  is nonsingular and hence  $\delta x = 0$ .  $\square$

**Remark 3.** Theorem 1 was proven in Pannocchia and Rawlings (2003) by using a different approach.

**Remark 4.** Theorem 1 can be extended to prove local Lyapunov stability of the closed-loop system (19) under standard regularity assumptions on the state update function  $f$  in (19) (Mayne et al., 2000).

**Remark 5.** The proof of Theorem 1 assumes only that the models used for the control design (3) and the observer design (4) are identical in steady state in the sense that they give rise to the same relation  $z = z(u, d, r)$ . It does not make any assumptions about the behavior of the real plant (1), i.e. the model-plant mismatch, with the exception that the closed-loop system (19) must converge to a fixed point. The models used in the controller and the observer could even be different as long as they satisfy the same steady state relation.

**Remark 6.** If condition (16) does not specify  $\bar{x}_t$  and  $\bar{u}_t$  uniquely, it is customary to determine  $\bar{x}_t$  and  $\bar{u}_t$  through an optimization problem, for example, minimizing the magnitude of  $\bar{u}_t$  subject to the constraint (16) (Pannocchia & Rawlings, 2003).

#### 4. Number of disturbance states $n_d$ different from number of measured outputs $p$

This section deals with the case when the number of measured variables  $p$  is greater than the chosen number of disturbance states  $n_d$ . If only a few measured variables are to be tracked without offset, choosing  $n_d = p$  as in the previous section, might introduce a possibly large number of disturbance states which in turn increases the complexity of the MPC problem.

From the internal model principle, it is clear that we have to add at least one disturbance state for every output to be tracked without offset. Hence, the number of additional disturbances  $n_d$  is chosen to be equal to the number of tracked outputs, i.e.,  $n_d = r < p$ . This choice, in general, does not guarantee offset-free tracking when the controller is designed as in Section 3, since Proposition 3 does not hold. In the following, we first derive a characterization of the tracking error at steady state. Then, we present a method for constructing the observer such that offset-free tracking is obtained.

##### 4.1. Steady-state tracking error

The controller (15)–(16) and observer (8) remain unchanged. However, since  $n_d < p$ , Proposition 3 is replaced by the following proposition.

**Proposition 4.** The steady state of the observer (8) satisfies

$$\begin{bmatrix} A - I + L_x C & B \\ L_d C & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_\infty \\ u_\infty \end{bmatrix} = \begin{bmatrix} L_x y_{m,\infty} - (B_d + L_x C_d) \hat{d}_\infty \\ L_d y_{m,\infty} - L_d C_d \hat{d}_\infty \end{bmatrix} \quad (25)$$

where  $y_{m,\infty}$  and  $u_\infty$  are the steady state output and input of the system (1), respectively,  $\hat{x}_\infty$  and  $\hat{d}_\infty$  are state and disturbance estimates from the observer (8) at steady state, respectively.

**Proof.** From (8) at steady state we obtain

$$\begin{aligned} \hat{x}_\infty &= A \hat{x}_\infty + B u_\infty + L_x (-y_{m,\infty} + C \hat{x}_\infty + B_d \hat{d}_\infty) \\ 0 &= L_d (-y_{m,\infty} + C \hat{x}_\infty + C_d \hat{d}_\infty). \end{aligned} \quad (26)$$

Eq. (25) follows directly from (26).  $\square$

In the proof of Theorem 1 we have shown that at steady state, the MPC controller (15)–(16) satisfies

$$u_\infty - \bar{u}_\infty = K_{\text{MPC}} (\hat{x}_\infty - \bar{x}_\infty), \quad (27)$$

where  $K_{\text{MPC}}$  is the unconstrained MPC (15)–(16). By combining the Eqs. (16), (25), (27) and using the offset equation  $\epsilon = H y_{m,\infty} - r_\infty$  we find that the observer and controller steady state values satisfy the following equation:

$$M_1 v + M_2 \epsilon = 0 \quad (28)$$

with  $v = [\hat{x}_\infty^T u_\infty^T \bar{x}_\infty^T \bar{u}_\infty^T \hat{d}_\infty^T y_{m,\infty}^T]^T$  and

$$M_1 = \begin{bmatrix} A + L_x C - I & B & 0 & 0 & B_d + L_x C_d & -L_x \\ L_d C & 0 & 0 & 0 & L_d C_d & -L_d \\ 0 & 0 & A - I & B & B_d & 0 \\ 0 & 0 & H C & 0 & H C_d & -H \\ K_{\text{MPC}} & -I & -K_{\text{MPC}} & I & 0 & 0 \end{bmatrix}$$

$$M_2 = [0 \ 0 \ 0 \ 0]^T.$$

The set of all possible offsets  $\epsilon$  belongs to the projection of the subspace  $\mathcal{K} \triangleq \{(v, \epsilon) | M_1 v + M_2 \epsilon = 0\}$  on the offset space, this is denoted by  $\Pi_\epsilon(\mathcal{K})$ .

The following Algorithm 4.1 describes a standard procedure to determine the smallest subspace  $\mathcal{H} = \Pi_\epsilon(\mathcal{K}) \subseteq \mathbb{R}^r$  spanned by all possible offsets  $\epsilon$ .

**Algorithm 4.1.** Step 1. **if**  $\text{rank}(M_1) = \text{rank}([M_1 \ M_2])$  then  $\Pi_\epsilon(\mathcal{K})$  is full dimensional, i.e.,  $\mathcal{H} \leftarrow \mathbb{R}^{n_d}$ ; **end**  
Step 2. **if**  $\text{rank}([M_1 \ M_2(:, 1 : j-1) \ M_2(:, j+1 : n_d)]) \neq \text{rank}([M_1 \ M_2])$  for all  $j = 1, \dots, n_d$  then we have zero steady state offset and  $\mathcal{H} \leftarrow \mathbf{0}$ .  
Step 3. **else** let  $U = [u_1, \dots, u_k]$  span the kernel of  $M_1^T$ . Then  $\mathcal{H} \leftarrow \{\epsilon | Z \epsilon = 0\}$ , where  $Z = U^T M_2$ ; **end**.

We emphasize that the synthesis of MPC controllers with zero steady state offset is not an easy problem. In fact,  $Z$  (and thus  $\mathcal{H}$ ) is a function of  $L_x$ ,  $L_d$ ,  $H$  and  $K_{\text{MPC}}$ . In the next section we provide an algorithm for computing  $L_x$ ,  $L_d$  such that zero steady state offset is obtained. First we make two important remarks.

**Remark 7.** The case when the  $p - r$  measured variables which are not tracked are discarded from the observer design:

$$\begin{aligned} \begin{bmatrix} \hat{x}(t+1) \\ \hat{d}(t+1) \end{bmatrix} &= \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}(t) \\ \hat{d}(t) \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u(t) \\ &\quad + \begin{bmatrix} L_x \\ L_d \end{bmatrix} H (-y(t) + C \hat{x}(t) + C_d \hat{d}(t)) \end{aligned} \quad (29)$$



falls in the class of systems studied in Section 2.3. Therefore, the conditions presented in this section are relevant only if all the measurements  $y$  are used for the observer design and  $r < p$ .

**Remark 8.** By defining  $e_\infty = y_{m,\infty} - C\hat{x}_\infty - C_d\hat{d}_\infty$ , (28) can be rewritten as follows

$$\tilde{M}_1 \tilde{v} + M_2 \epsilon = 0 \quad (30)$$

with  $\tilde{v} = [\hat{x}_\infty^T u_\infty^T \tilde{x}^T \hat{d}_\infty^T e_\infty^T]^T$  and

$$\tilde{M}_1 = \begin{bmatrix} A - I & B & 0 & 0 & B_d & -L_x \\ 0 & 0 & 0 & 0 & 0 & -L_d \\ 0 & 0 & A - I & B & B_d & 0 \\ -HC & 0 & HC & 0 & 0 & -H \\ K_{MPC} & -I & -K_{MPC} & I & 0 & 0 \end{bmatrix}. \quad (31)$$

By using direct substitution, the following equation can be derived from Eq. (30)

$$\left[ H(I - C(I - A - BK_{MPC})^{-1}L_x) \right] e_\infty = \begin{bmatrix} 0 \\ I \end{bmatrix} \epsilon. \quad (32)$$

Clearly Algorithm 4.1 can be applied to Eq. (32) as well.

From Eq. (32) we conclude that zero steady state offset is obtained if  $\epsilon = 0$  for all  $e_\infty$  solving (32), i.e., if  $H(I - C(I - A - BK_{MPC})^{-1}L_x)\epsilon_\infty = 0$  for all  $e_\infty$  satisfying  $L_d e_\infty = 0$ . This can be rewritten in the following null space condition

$$\mathcal{N}(L_d) \subseteq \mathcal{N}(H(I - C(I - A - BK_{MPC})^{-1}L_x)). \quad (33)$$

Eq. (33) is the main equation used in Pannocchia and Rawlings (2003).

#### 4.2. Algorithm for offset-free tracking when $n_d = r < p$

In the following, we propose a method for constructing  $L_x$  and  $L_d$  when  $n_d = r < p$  such that condition (33) holds. We assume that the MPC is defined as in Eqs. (15) and (16), and the unconstrained MPC controller gain  $K_{MPC}$  is given. We introduce the following notation for brevity

$$\Phi = I - A - BK_{MPC}, \quad (34)$$

and

$$A_m = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix}, \quad C_m = [C \ C_d]. \quad (35)$$

**Theorem 2.** Consider the augmented system model (4) with  $n_d = r$  and the estimator with a gain of the form

$$L = \begin{bmatrix} L_x \\ 0 \end{bmatrix} + \begin{bmatrix} \bar{L}_x \\ \bar{L}_d \end{bmatrix} \bar{H} \quad (36)$$

where  $\bar{H} = H(I - C\Phi^{-1}L_x)$ . Assume the closed-loop observer dynamics  $A_m + LC_m$  is stable. Then, controller (15)–(18) yields offset-free tracking.

**Proof.** Substituting (36) into (33) yields

$$\mathcal{N}(\bar{L}_d \bar{H}) \subseteq \mathcal{N}(H(I - C\Phi^{-1}(L_x + \bar{L}_x \bar{H}))),$$

$$\mathcal{N}(\bar{L}_d \bar{H}) \subseteq \mathcal{N}(\bar{H} - HC\Phi^{-1}\bar{L}_x \bar{H}),$$

$$\mathcal{N}(\bar{L}_d \bar{H}) \subseteq \mathcal{N}((I - HC\Phi^{-1}\bar{L}_x)\bar{H}).$$

The last inclusion holds true since  $\bar{L}_d$  is of full row rank as established in Proposition 2.  $\square$

**Remark 9.** A simple choice in (36) is to set  $L_x = 0$  and thus  $\bar{H} = H$ . This is clearly equivalent to the case discussed in Remark 7, since the number of measurements used by the observer  $y'_m(k) = Hy_m(k) = z(k)$  is equal to the number of disturbances. Next we are interested in the case when  $L_x \neq 0$  since neglecting measurements might negatively affect the observer performance.

Theorem 2 suggest a direct construction method for the estimator.

**Algorithm 4.2.** Consider the linear system (4) with the definitions (35). Assume  $(C_m, A_m)$  detectable. Suppose  $K_{MPC}$  is given.

Step 1. Choose  $L_x$  such that  $A + L_x C$  is stable and  $(\bar{H}C_m, \bar{A})$  detectable, where  $\bar{H} = H(I - C\Phi^{-1}L_x)$  and  $\bar{A} = A_m - [L_x^T \ 0]^T C_m$ .

Step 2. Choose  $\bar{L}$  such that  $\bar{A} - \bar{L}\bar{H}C_m$  is stable with  $\bar{L}^T = [\bar{L}_x^T \ \bar{L}_d^T]^T$ .

Step 3. Choose the final estimator gain

$$L = \begin{bmatrix} L_x \\ 0 \end{bmatrix} + \begin{bmatrix} \bar{L}_x \\ \bar{L}_d \end{bmatrix} \bar{H}.$$

**Remark 10.** The construction of the estimator gain  $L$  in (36) can be nicely interpreted when  $\bar{L}_x$  is equal to the identity matrix. In this case, during the transient  $L_x$  is used to generate a state estimation which is based on all measurements. At steady state, the corrective term  $\bar{H}$  cancels the effect of  $L_x$  and the steady state estimation relies only on the tracked measurement (which guarantees zero offset, see Remark 9). In order to demonstrate the last point, consider Eq. (26). Then,  $\hat{z}_\infty = HC\hat{x}_\infty = HC\Phi^{-1}\tilde{L}_x\epsilon_\infty$  since  $u_\infty = K_{MPC}\hat{x}_\infty$ . Choosing  $\tilde{L}_x = L_x + \bar{H}$  and  $\bar{H} = H(I - C\Phi^{-1}L_x)$  we obtain  $\hat{z}_\infty = HC\Phi^{-1}L_x\epsilon_\infty - HC\Phi^{-1}L_x\epsilon_\infty + HC\Phi^{-1}H\epsilon_\infty$ . Therefore at steady state  $\hat{z}_\infty = HC\Phi^{-1}H\epsilon_\infty$ .

An alternative way to construct the estimator is described in the following modified algorithm, which allows one to move the closed-loop poles associated with the disturbance estimates independently from the observed states modes.

**Algorithm 4.3.** Step 1. Choose  $L_x$  such that  $A - L_x C$  is stable and  $(\bar{H}C_m, \bar{A})$  detectable, where  $\bar{H} = H(I + C\Phi^{-1}L_x)$  and  $\bar{A} = A_m - [L_x^T \ 0]^T C_m$ .

Step 2. Apply the linear transform

$$T = \begin{bmatrix} I & -(I - A - L_x C)^{-1}(B_d + L_x C_d) \\ 0 & I \end{bmatrix}$$

which brings the system to block-diagonal form

$$\bar{A}_t = T\bar{A}T^{-1} = \begin{bmatrix} A + L_x C & 0 \\ 0 & I \end{bmatrix},$$

$$\begin{aligned} \bar{C}_t &= \bar{H}C_m T^{-1} \\ &= \bar{H}[C \ C_d(I - A - L_x C)^{-1}(B_d + L_x C_d) + C_d]. \end{aligned}$$

Step 3. Choose  $\bar{L}_d$  such that

$$I - \bar{L}_d \bar{H}[C(I - A - L_x C)^{-1}(B_d + L_x C_d) + C_d]$$

is stable.

Step 4. Compute the estimator gain for the original system

$$L = \begin{bmatrix} L_x \\ 0 \end{bmatrix} + T^{-1} \begin{bmatrix} 0 \\ \bar{L}_d \bar{H} \end{bmatrix}.$$

**Remark 11.** If  $L$  is computed as in Algorithm 4.3, then the closed-loop poles of the estimator are the eigenvalues of  $A - L_x C$  and the eigenvalues of  $I - \bar{L}_d \bar{H}[C(I - A - L_x C)^{-1}(B_d + L_x C_d) + C_d]$ . They can be assigned independently.

**Remark 12.** Algorithms 4.2 and 4.3 require that  $L_x$  can be chosen such that  $(\bar{H}C_m, \bar{A})$  is detectable. From Algorithm 4.3, it is clear that detectability holds if and only if

$$\bar{H}[C(I - A - L_x C)^{-1}(B_d + L_x C_d) + C_d] \quad (37)$$

is full row rank. We note that this is not a restrictive assumption. By noting that

$$\bar{H}C = HC\Phi^{-1}(I - A - BK_{MPC} - L_x C) \quad (38)$$

we observe two cases where (37) might lose rank:

- (1) If  $(I - A - BK_{MPC} - L_x C)$  is not full rank and  $\text{span}(HC\Phi^{-1}) \cap \mathcal{N}((I - A - BK_{MPC} - L_x C)^T) \neq \{0\}$ , then (37) loses rank. Because of the degree of freedom we have in choosing  $K_{MPC}$  and  $L_x$ , we may safely assume that this rank deficiency can be avoided.
- (2) If  $\text{span}((I - A - L_x C)^{-1}(B_d + L_x C_d) + C_d) \cap \mathcal{N}(\bar{H}C) \neq \{0\}$ , then (37) loses rank. In this case, either the disturbance model  $B_d, C_d$  or the estimator gain  $L_x$  can be modified in order for (37) to have full rank.

Note that Algorithms 4.2 and 4.3 also require that  $L_x$  is chosen such that  $A - L_x C$  is stable. Conditions on  $A, C, C_m, \bar{H}, \bar{A}$  guaranteeing both stability and a detectability properties are subject of current study.

**Remark 13.** In Algorithm 4.3, the matrix  $\bar{H}$  depends on  $\Phi$  and thus on the controller gain  $K_{MPC}$ . Assume that Algorithm 4.3 has been executed for a given MPC tuning yielding  $\bar{L}_d$ . If the same MPC is redesigned with a different tuning with corresponding  $K_{MPC,1}$  and  $\bar{H}_1$ , then in Step 3 of Algorithm 4.3 one can choose

$$\bar{L}_{d,1} = (I - A)(\bar{H}_1[C(I - A - L_x C)^{-1}(B_d + L_x C_d) + C_d])^{-1} \quad (39)$$

where

$$\Lambda = I - \bar{L}_d \bar{H}[C(I - A - L_x C)^{-1}(B_d + L_x C_d) + C_d]. \quad (40)$$

This will ensure identical observer performance regardless of the controller used. Note that for  $\bar{L}_{d,1}$  to exist, the detectability condition as discussed in Remark 12 needs to hold.

## 5. Special MPC classes

### 5.1. Different norm in the objective function

If the 2-norm in the objective function of (15) is replaced with a 1 or  $\infty$  norm ( $\|P(x_N - \bar{x}_t)\|_p + \sum_{k=0}^{N-1} \|Q(x_k - \bar{x}_t)\|_p + \|R(u_k - \bar{u}_t)\|_p$ , where  $p = 1$  or  $p = \infty$ ), then the results of Section 3 continue to hold. In particular, Theorem 1 continues to hold. In fact, the unconstrained MPC controlled  $K_{MPC}$  in (23) is piecewise linear around the origin (Bemporad, Morari, Dua, & Pistikopoulos, 2002). In particular, around the origin,  $\delta u^*(t) \triangleq \delta u_0^* = K_{MPC}(\delta x(t))$  is a continuous piecewise linear function of the state variation  $\delta x$ :

$$K_{MPC}(\delta x) = F^i \delta x \quad \text{if } H^i \delta x \leq K^i, i = 1, \dots, N^r \quad (41)$$

where  $H^i$  and  $K^i$  in Eq. (41) are the matrices describing the  $i$ th polyhedron  $CR^i = \{\delta x \in \mathbb{R}^n | H^i \delta x \leq K^i\}$  inside which the feedback optimal control law  $\delta u^*(t)$  has the linear form  $F^i \delta x(k)$ . The polyhedra  $CR^i, i = 1, \dots, N^r$  are a partition of the set of feasible states of problem (15) and they all contain the origin. For Theorem 1 to hold, it is sufficient to require that all the linear feedback laws  $F^i \delta x(k)$  for  $i = 1, \dots, N^r$  are stabilizing. For the case  $n_d = r < p$ , condition (33) extends to

$$\mathcal{N}(L_d) \subseteq \bigcap_{i=1}^{N_r} \mathcal{N}(H(I - C(I - A - BF^i)^{-1}L_x)). \quad (42)$$

As a consequence, Algorithms 4.2 and 4.3 cannot be applied directly. Therefore, a full disturbance model with  $n_d = p$  has to be chosen for 1 and  $\infty$  norms.

### 5.2. Explicit controller

In the last few years there has been growing interest to apply MPC to systems where the computational resources are insufficient to solve the optimization problem (15) and (16) on-line in real time. Methods have been developed (Bemporad, Borrelli, & Morari, 2002; Borrelli, 2003; Borrelli, Baotic, Bemporad, & Morari, 2005; Kvasnica, Grieder, & Baotic, 2004) to solve (15) and (16) explicitly to obtain a state feedback control law  $u(t) = c_0(\hat{x}(t), \hat{d}(t), r(t))$  in the form of a look-up table. The applicability of these methods is limited by the complexity of the control law  $c_0(\cdot)$  which is greatly affected by the number of parameters, e.g. the number of elements in the vectors  $\hat{x}(t), \hat{d}(t)$  and  $r(t)$ .

Thus it is of interest to examine the proposed control formulations and disturbance models from this perspective. Examining (15) and (16) we note that the control law depends on  $\hat{x}(t), \hat{d}(t)$  and  $r(t)$ . For instance,  $n_d = p$  is a popular choice of disturbance model dimension, since it yields offset-free control by default, as was seen in Section 3. However, this leads to  $p + r$  additional parameters. In Section 4 a method was proposed to obtain offset-free control for models with  $n_d < p$  and in particular, with minimum order disturbance models  $n_d = r$ . The total size of the parameter vector can thus be reduced to  $n + 2r$ . This is significant only if a small subset of the plant outputs are to be controlled.

A greater reduction of parameters can be achieved by the following method. By Corollary 1, we are allowed to choose  $B_d = 0$  in the disturbance model if the plant has no integrators. Recall the target conditions (16) with  $B_d = 0$

$$\begin{bmatrix} A - I & B \\ HC & 0 \end{bmatrix} \begin{bmatrix} \bar{x}_t \\ \bar{u}_t \end{bmatrix} = \begin{bmatrix} 0 \\ r(t) - HC_d \hat{d}(t) \end{bmatrix}. \quad (43)$$

Clearly, any solution to (43) can be parameterized by  $r(t) - HC_d \hat{d}(t)$ . The explicit control law is written as  $u(t) = c_0(\hat{x}(t), r(t) - HC_d \hat{d}(t))$ , with  $n + r$  parameters. Since the observer is unconstrained, complexity is much less of an issue. Hence, a full disturbance model with  $n_d = p$  can be chosen, yielding offset-free control by default.

**Remark 14.** The choice of  $B_d = 0$  might be limiting in practice. In Vibhor and Borrelli (2008), the authors have shown that for a wide range of systems, if  $B_d = 0$  and the observer is designed through a Kalman filter, then the closed loop system might suffer a dramatic performance limitation.

### 5.3. Delta input ( $\delta u$ ) formulation

In  $\delta u$  formulation, the MPC scheme uses of the following linear time-invariant system model of (1):

$$\begin{cases} x(t+1) = Ax(t) + Bu(t) \\ u(t) = u(t-1) + \delta u(t) \\ y(t) = Cx(t). \end{cases} \quad (44)$$

System (44) is controllable if  $(A, B)$  is controllable. The  $\delta u$  formulation often arises naturally in practice when the actuator is subject to uncertainty, e.g. the exact gain is unknown or is subject to drift. In these cases, it can be advantageous to consider changes in the control value as input to the plant. The absolute control value is estimated by the observer, which is expressed as follows

$$\begin{bmatrix} \hat{x}(t+1) \\ \hat{u}(t+1) \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}(t) \\ \hat{u}(t) \end{bmatrix} + \begin{bmatrix} B \\ I \end{bmatrix} \delta u(t) + \begin{bmatrix} L_x \\ L_u \end{bmatrix} (-y_m(t) + C\hat{x}(t)). \quad (45)$$

The MPC problem is readily modified

$$\begin{aligned} \min_{\delta u_0, \dots, \delta u_{N-1}} \quad & \|y_k - r_k\|_Q^2 + \|\delta u_k\|_R^2 \\ \text{subj. to} \quad & Ex_k + Lu_k \leq M, \quad k = 0, \dots, N-1 \\ & x_{k+1} = Ax_k + Bu_k, \quad k \geq 0 \\ & y_k = Cx_k, \quad k \geq 0 \\ & u_k = u_{k-1} + \delta u_k, \quad k \geq 0 \\ & u_{-1} = \hat{u}(t) \\ & x_0 = \hat{x}(t). \end{aligned} \quad (46)$$

The control input applied to the system is

$$u(t) = \delta u_0^* + u(t-1). \quad (47)$$

The input estimate  $\hat{u}(t)$  is not necessarily equal to the actual input  $u(t)$ . This scheme inherently achieves offset-free control, there is no need to add a disturbance model. To see this, we first note that  $\delta u_0^* = 0$  in steady-state. Hence, the analysis presented in Section 3 applies as the  $\delta u$  formulation is equivalent to a disturbance model in steady-state. This is due to the fact that any plant/model mismatch is lumped into  $\hat{u}(t)$ . Indeed this approach is equivalent to an input disturbance model ( $B_d = B$ ,  $C_d = 0$ ). If in (46) the measured  $u(t)$  is substituted to its estimate, i.e.  $u_{-1} = u(t-1)$ , then the algorithm would show offset.

In this formulation the computation of a target input  $\bar{u}_t$  and state  $\bar{x}_t$  is not required. A disadvantage of the formulation is that it is not applicable when there is an excess of manipulated variables in  $u$  compared to measured variables  $y$ , since detectability of the augmented system (44) is lost.

#### 5.4. Minimum-time controller

In minimum-time control, the cost function minimizes the predicted number of steps necessary to reach a target region, usually the invariant set associated to the unconstrained LQR controller (Keerthi & Gilbert, 1987). This scheme can reduce the on-line computation time significantly, especially for explicit controllers (Grieder & Morari, 2003). While minimum-time MPC is computed and implemented differently from standard MPC controllers, there is no difference between the two control schemes at steady-state. In particular, one can choose the target region to be the unconstrained region of (15). When the state and disturbance estimates and reference are within this region, the control law is switched to (15). The analysis and methods presented in this paper therefore apply directly.

## 6. Examples

In this section, two examples are discussed. The purpose of the first one is to illustrate the anti-windup effect of the proposed controller, while the second example shows the application of Algorithm 4.3.

### 6.1. Integral action and anti-windup

Consider the system

$$\begin{aligned} x(t+1) &= ax(t) + u(t) + d(t), \\ y(t) &= x(t) \end{aligned} \quad (48)$$

with input constraints:

$$|u(t)| \leq 1. \quad (49)$$

The reference value is assumed to be constant and equal to 0. The goal is to design a controller which achieves zero offset, i.e.  $y(t) \rightarrow 0$  as  $t \rightarrow \infty$ .

The MPC is formulated as follows

$$\begin{aligned} \min_{u_0} \quad & (u_0 - \bar{u}_t)^2 + (x_1 - \bar{x}_t)^2 \\ \text{subj. to} \quad & -1 \leq u_{0,t} \leq 1 \\ & x_1 = a\hat{x}(t) + u_{0,t} + \hat{d}(t) \\ & \begin{bmatrix} a-1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \bar{x}_t \\ \bar{u}_t \end{bmatrix} = \begin{bmatrix} -\hat{d}(t) \\ 0 \end{bmatrix}. \end{aligned} \quad (50)$$

The closed form solution to (50) can be easily computed:

$$u^*(t) = \begin{cases} 1, & -\hat{d}(t) - \frac{1}{2}a\hat{x}(t) > 1, \\ -1, & -\hat{d}(t) - \frac{1}{2}a\hat{x}(t) < -1, \\ -\hat{d}(t) - \frac{a}{2}\hat{x}(t) & \text{otherwise.} \end{cases} \quad (51)$$

Note that  $K_{\text{MPC}} = -\frac{a}{2}$ .

Since the number of measured variables equals the number of disturbances used in the model, any stabilizing observer will achieve offset-free control. We choose

$$L = -\begin{bmatrix} a \\ 1/4 \end{bmatrix}. \quad (52)$$

The dynamics of the controller in the unconstrained case is thus given by the piecewise affine system

$$\tilde{x}(t+1) = \begin{cases} \tilde{A}_c \tilde{x}(t) - Ly(t) + f, & h^T \tilde{x}(t) > 1, \\ \tilde{A}_c \tilde{x}(t) - Ly(t) - f, & h^T \tilde{x}(t) < -1, \\ \tilde{A}_u \tilde{x}(t) - Ly(t), & \text{otherwise} \end{cases} \quad (53)$$

with

$$\tilde{A}_u = \begin{bmatrix} \frac{a}{2} & 0 \\ -\frac{1}{4} & 1 \end{bmatrix}, \quad \tilde{A}_c = \begin{bmatrix} 0 & 1 \\ -\frac{1}{4} & 1 \end{bmatrix}, \quad (54)$$

$$f = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad h = -\begin{bmatrix} \hat{d}(t) \\ \frac{1}{2}a\hat{x}(t) \end{bmatrix} \quad (55)$$

and  $\tilde{x}(t) = [\hat{x}(t)^T \hat{d}(t)^T]^T$ .

One can notice that the unconstrained dynamics  $\tilde{A}_u$  contains an integrator, while the constrained dynamics  $\tilde{A}_c$  is asymptotically stable (two poles in 0.5). Hence, when the system saturates, we obtain an anti-windup effect.

### 6.2. Multivariable system

Consider the linearized airplane model discussed in Maciejowski (1998)

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} 0 & 0 & 1.132 & 0 & -1.0 \\ 0 & -0.054 & -0.171 & 0 & 0.071 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0.049 & 0 & -0.856 & -1.01 \\ 0 & -0.291 & 0 & 1.053 & -0.686 \end{bmatrix} x(t) \\ &+ \begin{bmatrix} 0 & 0 & 0 \\ -36 & 20 & 0 \\ 0 & 0 & 0 \\ 1325 & 0 & -57 \\ 472 & 0 & -3 \end{bmatrix} u(t), \end{aligned} \quad (56)$$

$y_m(t) = [I_{3 \times 3} \ 0_{3 \times 2}]x(t)$ .

The state vector  $x = [x_1, \dots, x_5]$  comprises altitude, horizontal speed, pitch angle, pitch rate and vertical speed, respectively. The input variables  $u_1$ ,  $u_2$  and  $u_3$  are spoiler deflection, engine thrust and elevator angle, respectively. The inputs are constrained as follows:  $-1 \leq u_i(t) \leq 1$ .

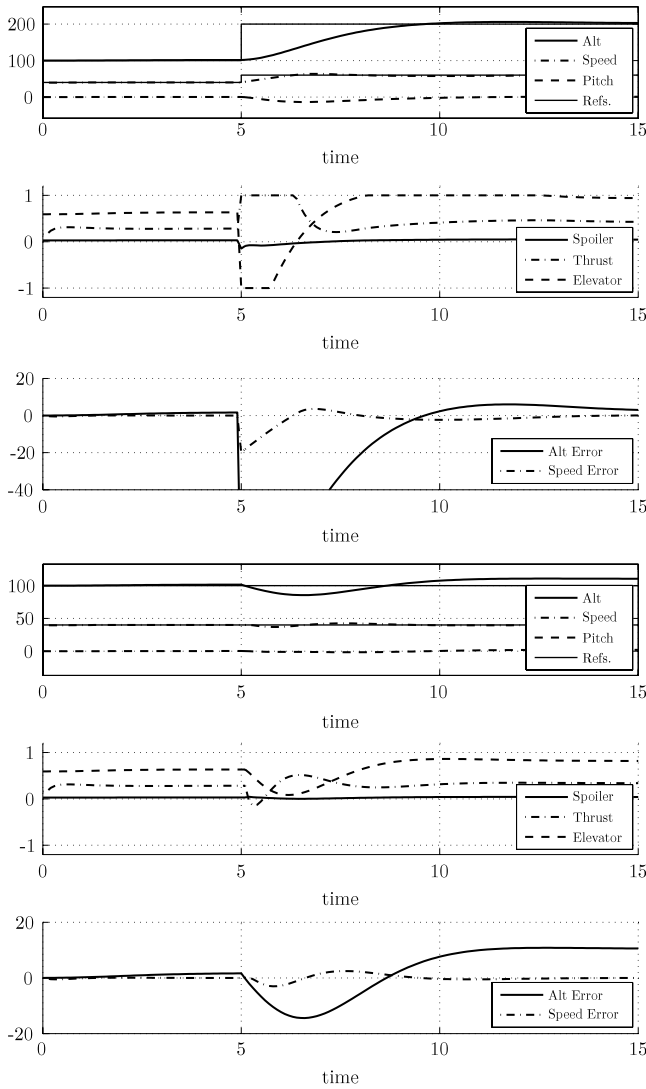


Fig. 1. Reference and disturbance step responses.

The continuous-time model (56) is discretized with a sampling period of  $T_s = 0.1$ . We design an MPC controller in order to track altitude and horizontal speed, hence  $H = [I_{2 \times 2} \ 0_{2 \times 3}]$ , but zero offset is required on the speed only. According to the results presented in Section 4.2 a disturbance model with  $n_d = 1$  is sufficient for obtaining zero steady-state offset.

The MPC is posed as in (15) and (16) with  $Q = \text{diag}([100 \ 100 \ 100 \ 1 \ 1])$ ,  $R = 10^4 \cdot \text{diag}([1 \ 1 \ 10])$  and a prediction horizon of  $N = 3$ . The resulting feedback gain depends on  $n + r + n_d = 8$  parameters. The estimator Algorithm 4.2 is employed, where the gains  $L_x$  and  $\bar{L}$  are the solutions of the steady-state Kalman filter with unitary weights. Since only one variable is to be controlled without offset, in Algorithm 4.2 we set  $\bar{H} = [0 \ 1 \ 0 \ 0 \ 0](I - C\Phi^{-1}L_x)$ .

Fig. 1 depicts two simple tests which show offset free control on the horizontal speed. In both tests we simulate a model mismatch by doubling the drag coefficient (i.e., multiply by two the element (2, 2) in the matrix  $A$  of model (56)) and reducing all the actuator gains by 5%. In the first test (left section of Fig. 1) we simulate a step change in the reference at time  $t = 5$  s. In the second test (right section of Fig. 1) we simulate an additive disturbance at time  $t = 5$  s. The disturbance represents a wind gust with horizontal and vertical components (headwind and downdraft).

## 7. Conclusion

We discussed the problem of offset-free Model Predictive Control when tracking an asymptotically constant reference. The system was augmented by additional disturbance states, and a linear disturbance observer was employed to obtain disturbance estimates. Simple conditions for zero offset have been derived from a steady-state analysis of both the estimator and controller. We first treated the case when the number of disturbances is equal to the number of measured variables ( $n_d = p$ ), which yields zero-offset in a straightforward way. This approach may, however, introduce more disturbance states than there are controlled variables, and thus lead to more complex MPC problems than necessary.

Then, the case when  $n_d = r < p$  was discussed, which does in general not yield zero offset (Muske & Badgwell, 2002; Pannocchia, 2003; Pannocchia & Rawlings, 2003). We have proposed an algorithm for computing the observer in such a way that the offset is removed in selected variables. Thus, the resulting controller has fewer parameters and is less complex than with previous methods.

Insights were given into the important cases when the performance objective is a 1 or  $\infty$  norm, and when the MPC is computed explicitly to reduce on-line computation time.

We remark that Reference Governor (RG) algorithms (Bemporad, 1998; Bemporad, Casavola, & Mosca, 1997; Bemporad & Mosca, 1994a,b, 1995) provide an alternative, attractive way for designing tracking controllers for constrained systems. Since RG make use of a prediction model of the closed loop (plant + controller), the results presented in this work can be easily applied to RG design.

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